

Semester: IV
Section: IT,CSIT,ME,EN,ECE,CE
Sub. Code: BOE-404
Time: 3 hrs.

OBE Remarks:																	
Q.No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
CO No.	1	2	3	5	3	4	5	1	2	1	3	3	3	4	4	5	5
Bloom's Level	L1	L2	L2	L2	L2	L2	L1	L2	L2	L3	L4	L3	L2, L4	L2, L4	L3, L5	L3, L2	L4, L2
Weightage CO3: 16					Weightage CO4: 16					Weightage CO5: 16							

- Explain the fundamentals of solar radiation. What are the devices used for measuring the solar radiations? Explain with any one of them with neat sketch.
- Explain the difference between direct radiation and diffuse radiation with the help of suitable diagram.

Roll No.

14. Attempt any one.

- a) What do you mean by dry system and wet steam geothermal systems? Explain with suitable diagram.
- b) Explain open cycle and closed cycle OTEC system with suitable diagram. State the limitations of OTEC System.

15. Attempt any one.

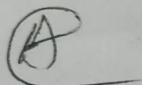
- a) Describe wind turbine design and dynamics with a neat diagram.
- b) Write a detailed note on the challenges and future scope of non-conventional energy in India.

16. Attempt any one.

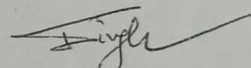
- a) Describe the process and importance of energy audit of industrial and commercial facilities.
- b) Explain the importance of energy conservation in engineering and industry.

17. Attempt any one.

- a) Propose methods for reducing the carbon footprint of a small city. Include energy conservation techniques, renewable energy sources, and waste management solutions.
- b) Explain the following concepts in detail: (i). Climate Change (ii). LEED Ratings.



Faculty Sign



HoD Sign